

AYMANE AIT DADS

Data Science Intern - Model Optimization | PyTorch · FP16 · Transformer Fine-Tuning

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Hiring Team - Data Science Intern - Model Optimization

quadric, Inc · San Jose, CA, USA

Dear Quadric Hiring Team,

Quadric's GPNPU is built to run **neural network inference workloads** efficiently across the full ML graph - a constraint that demands models be optimized before they ever touch silicon. That is exactly the problem I attacked when fine-tuning **CLIP ViT-L/14 (428M parameters)** on 250K images under a hard T4 GPU budget: I applied **FP16 mixed precision with GradScaler**, controlled transformer block unfreezing depth, and ranked **#1 on the EURECOM leaderboard** while keeping the model generalizable to out-of-distribution generators.

Optimizing for real hardware constraints is a thread across my work. In my ablation study on the CLIP model, I systematically varied **unfreezing depth (4, 6, and 8 transformer blocks)** and learning rate schedules, proving that **1-epoch fine-tuning** preserves generalization better than extended re-training - a finding directly relevant to fitting NN inference graphs within fixed edge-device budgets. At Orange Maroc, I engineered an end-to-end **ML pipeline over 100K+ telemetry records** using **Random Forest and KMeans**, reducing manual analysis time by **~60%** while mapping outputs to concrete operational decisions.

Quadric's bet on a general-purpose NPU that runs the entire ML graph - not just accelerable layers - is the architectural gap the rest of the industry has ignored, and it is the exact context where model-level optimization choices become the deciding factor for deployment viability. I would welcome the chance to discuss how my optimization and inference work fits the team's current roadmap.

Sincerely,

AYMANE AIT DADS